Tianyi Zhou

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Tianyi's Homepage
Tianyi's Google Scholar
Tianyi Lab Github

EDUCATION

University of Washington, Seattle
Doctor of Philosophy, Computer Science, Advisor: Prof. Jeff Bilmes 09/2013-08/2022

WORKING

Mohamed bin Zayed University of Artificial Intelligence, UAE 01/2026-present
Visiting Assistant Professor

University of Maryland, College Park, MD 09/2022-12/2025
Assistant Professor, Department of Computer Science

Google LLC, Seattle, WA 11/2021-08/2022
Visiting Research Scientist

RESEARCH FIELDS

Machine Learning, Natural Language Processing
- LLMs & Multi-modal Models, Mixture-of-Experts, Interactive Agents, Human-AI-World Alignment

PUBLICATION SUMMARY

1. Citations: 20116; 861(2022)→1480(2023)→3027(2024)→5418(2025)→6718(2026)
2. Most-cited paper: 1490, H-index: 65, I10-index: 182
3. World' Top 2% Scientists by Stanford in 2024, 2025, 2026 (self-citation ratio 5.18%, 5.57%, 4.35%).
4. 166 Conference Papers (20 NeurIPS, 22 ICML, 24 ICLR, 18 ACL, 12 EMNLP, 7 NAACL, 8 COLM, 5 AAAI, 5 IJCAI, 4 KDD, 3 ICCV, 4 CVPR, 4 ECCV)
5. 17 Journal Papers (IEEE TPAMI/TIP/TNNLS/TKDE, Machine Learning Journal, TMLR, WWW, DMKD)

HIGHLY CITED PUBLICATIONS

1. Tianrui Guan* and Fuxiao Liu*, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, and **Tianyi Zhou**. "*HallusionBench: An Advanced Diagnostic Suite for Entangled Language Hallucination & Visual Illusion in Large Vision-Language Models*", **CVPR**, 2024. (**1284 citations, 342 stars on Github**)
2. Yue Tan, Guodong Long, Lu Liu, **Tianyi Zhou**, Qinghua Lu, Jing Jiang, Chengqi Zhang, "*FedProto: Federated Prototype Learning across Heterogeneous Clients*", **AAAI**, 2022. (**1490 citations**)
3. Tao Shen†, **Tianyi Zhou**, Guodong Long, Jing Jiang, Shirui Pan, and Chengqi Zhang, "*DiSAN: Directional self-attention network for RNN/CNN-free language understanding*", **AAAI**, 2018. (**1011 citations, 312 Github stars, arXiv-ed 2 months after Transformer**)
4. **Tianyi Zhou** and Dacheng Tao, "*GoDec: Randomized Low-rank & Sparse Matrix Decomposition in Noisy Case*", **ICML**, 2011. (**954 citations**)

SELECTED PUBLICATIONS (2024-2026)

1. Ziyue Li, Yang Li, **Tianyi Zhou**, "*Skip a Layer or Loop it? Learning Program-of-Layers in Pretrained LLMs*", Forty-third International Conference on Machine Learning (**ICML**), 2026. (**Oral, 0.7%**)
2. Yuchen Wang, Jiangtao Kong, Sizhe Wei, Xiaochang Li, Haohong Lin, Hongjue Zhao, **Tianyi Zhou**, Lu Gan, Huajie Shao, "*WestWorld: A Knowledge-Encoded Scalable Trajectory World Model for Diverse Robotic Systems*", Forty-third International Conference on Machine Learning (**ICML**), 2026. (**Spotlight, 2.2%**)

3. Ming Li*, Chenrui Fan*, Yize Cheng*, Soheil Feizi, **Tianyi Zhou**, "*Schoenfeld's Anatomy of Mathematical Reasoning by Language Models*", Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026. (**Oral, SAC Highlight Award**)
4. Jiuhai Chen, Zhiyang Xu, Xichen Pan, Yushi Hu, Can Qin, Tom Goldstein, Lifu Huang, **Tianyi Zhou**, Saining Xie, Silvio Savarese, Le Xue, Caiming Xiong, Ran Xu, "*BLIP3-o: A Family of Fully Open Unified Multimodal Models-Architecture, Training and Dataset*", arXiv:2505.09568, 2025. (**363 citations, 1700+ stars on Github**)
5. Siyu Zhou, **Tianyi Zhou**, Yijun Yang, Guodong Long, Deheng Ye, Jing Jiang, Chengqi Zhang, "*WALL-E: World Alignment by NeuroSymbolic Learning improves World Model-based LLM Agents*", The Thirty-ninth Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
6. Ming Li, Yanhong Li, **Tianyi Zhou**, "*What Happened in LLMs Layers when Trained for Fast vs. Slow Thinking: A Gradient Perspective*", The 63rd Annual Meeting of the Association for Computational Linguistics (**ACL**), 2025. (**Oral, 2.9%**)
7. Ziyue Li, **Tianyi Zhou**, "*Your Mixture-of-Experts LLM Is Secretly an Embedding Model For Free*", The Thirteenth International Conference on Learning Representations (**ICLR**), 2025. (**Oral, 1.8%, 92 stars on Github**)
8. Ming Li, Yong Zhang, Shwai He, Zhitao Li, Hongyu Zhao, Jianzong Wang, Ning Cheng, **Tianyi Zhou**, "*Superfiltering: Weak-to-Strong Data Filtering for Fast Instruction-Tuning*", Annual Meeting of the Association for Computational Linguistics (**ACL**), 2024. (**200 citations, 189 stars on Github**)
9. Lichang Chen, Jiuhai Chen, Tom Goldstein, Heng Huang, and **Tianyi Zhou**. "*InstructZero: Efficient Instruction Optimization for Black-Box Large Language Models*", International Conference on Machine Learning (**ICML**), 2024. (**191 citations, 201 stars on Github**)
10. Yijun Yang, **Tianyi Zhou**, Kanxue Li, Dapeng Tao, Lusong Li, Li Shen, Xiaodong He, Jing Jiang, and Yuhui Shi. "*Embodied Multi-Modal Agent trained by an LLM from a Parallel TextWorld*", IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024. (**136 citations, 63 stars on Github**)

AWARDS & HONORS

1. Best Paper Honorable Mention, NeurIPS 2026 Workshop on AI-Native Academia.
2. SAC Highlight Award from ACL 2026.
3. Best Research Award from the International e-Assessment Association (eAA) 2026.
4. Top 10% of high-scoring reviewers for NeurIPS 2020.
5. Best Student Paper Award from International Conference on Data Mining (ICDM), 2013.

ADVISING

1. 14 Ph.D. students at the University of Maryland: 2 graduated, 8 in their fourth year, 4 in their second year, including **one Apple Scholar 2026 (Ming Li), one UMD Flagship Fellowship 2025 (Dongping Chen)**.
2. 4 master's students at the University of Maryland: 3 graduated.
3. 4 undergraduate students at the University of Maryland.
4. Mentored 20+ Ph.D. and master students at other universities in the US and Australia since 2016, including **a leading contributor to OpenAI 4o image generation (Lu Liu)**.

TEACHING

1. **CMSC 421 (Undergrad): Introduction to Artificial Intelligence**
   CMSC 421 is an introductory course in Artificial Intelligence (AI) for senior undergraduate students. The goal is to introduce a range of ideas and methods in AI, including automated heuristic search, planning, games, statistical inference, reinforcement learning, deep neural networks, large language models, and robotics. Programming projects and optional research projects will help students obtain hands-on practice on various topics. **I updated the curriculum design by introducing topics such as deep neural networks and large language models, and optional research projects** 2022/2023 Fall

2. **CMSC 828A (Grad): Fantastic Machine Learning Paradigms and Where to Use Them**
   CMSC 828A provides an in-depth tour of different machine learning paradigms, covering their assumptions, formulations, motivations, where to use them, and the ideas connecting them. The course follows three routes: (1) supervised learning → semi-supervised learning→ unsupervised and self-supervised learning → online learning and reinforcement learning; (2) Single-task learning → transfer learning → multi-task learning → meta-learning; (3) Single-model learning → adversarial learning → ensemble learning → federated learning → multi-agent learning. **I developed the whole course, project topics, and reading list from scratch** 2023/2024 Spring

INSTITUTIONAL SERVICE

1. Field Committee in Artificial Intelligence at UMD CS, 2023-2024.
2. Graduate Admissions Committee at UMD CS, 2023, 2025
3. Award Committee of 2024 Rising Stars in Machine Learning

PROFESSIONAL SERVICE

1. **Area Chair** for The Annual Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
2. **Area Chair** for International Conference on Machine Learning (ICML) 2026
3. **Area Chair** for International Conference on Learning Representations (ICLR) 2025, 2026
4. **Meta-reviewer (Area Chair)** for The Annual Meeting of the Association for Computational Linguistics (ACL) 2025, 2026
5. **Meta-reviewer (Area Chair)** for The Conference on Empirical Methods in Natural Language Processing (EMNLP) 2025, 2026
6. **Student Travel Award Chair** for The Web Conference (WWW) 2025
7. **Meta-reviewer (Area Chair)** for International Conference on Pattern Recognition (ICPR) 2024
8. **Meta-reviewer (Area Chair)** for International Joint Conference on Artificial Intelligence (IJCAI) 2023, 2025
9. **Area Chair (SPC)** for Winter Conference on Applications of Computer Vision (WACV) 2022, 2023, 2024
10. **Area Chair (SPC)** for ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD) 2022
11. **Area Chair (SPC)** for AAAI Conference on Artificial Intelligence (AAAI) 2021, 2026